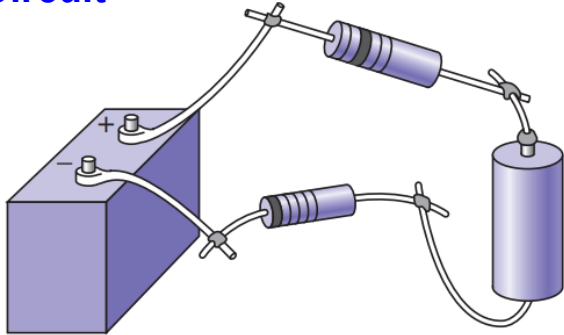


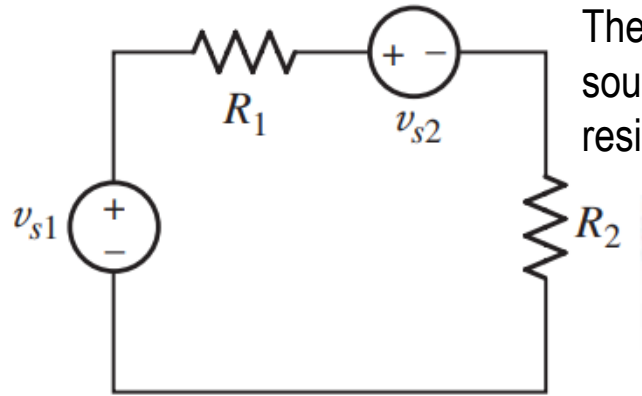
Series Circuit

All of the elements in a circuit that carry the **same current** are said to be connected in **series**.
A node can only have two elements connected it!!

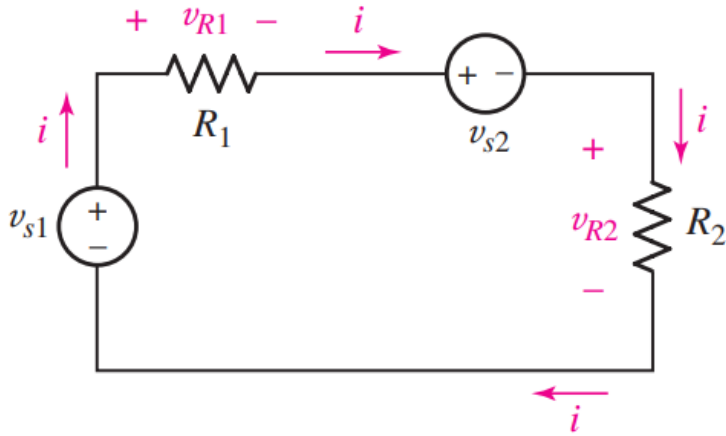
Real Circuit



(a)



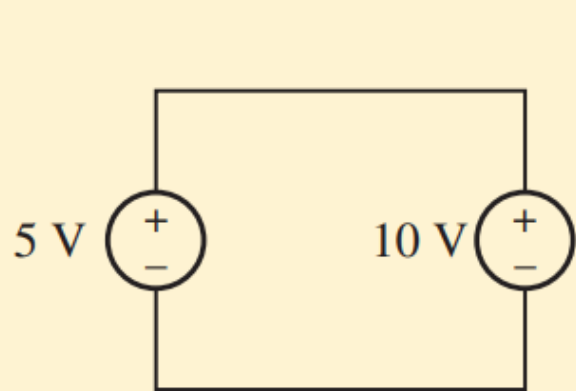
The circuit model with
source voltages and
resistance values given.



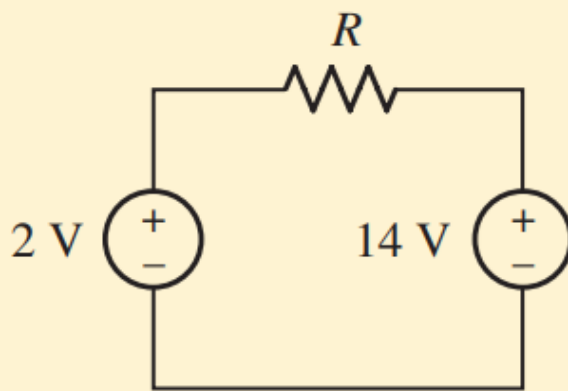
(c)

Analysis:

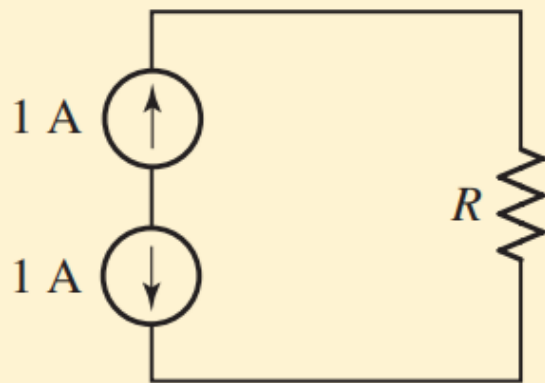
1. Assume reference current direction.
2. Apply KCL: same current.
3. Select voltage ref.
4. Apply KVL:



(a)



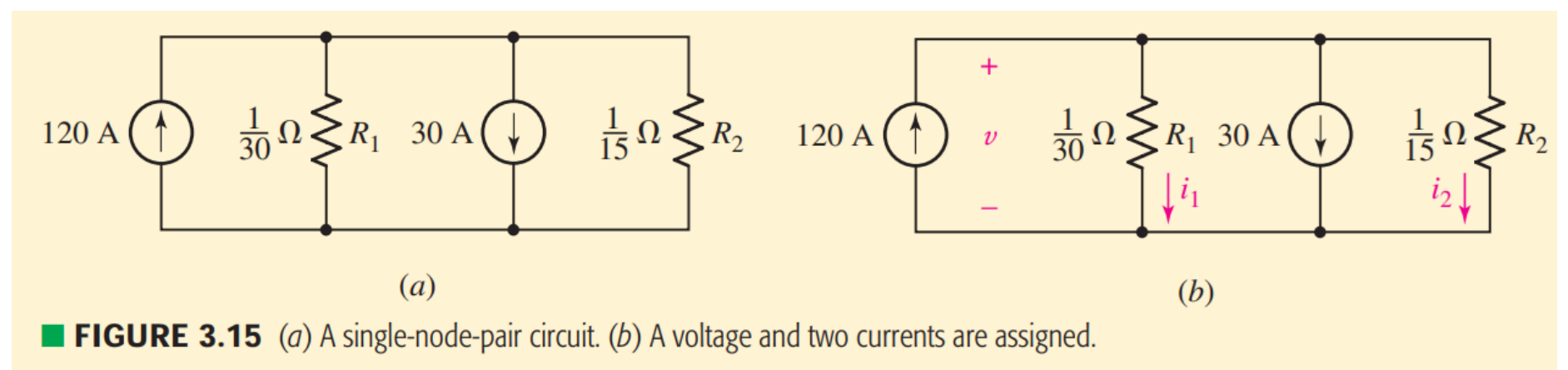
(b)

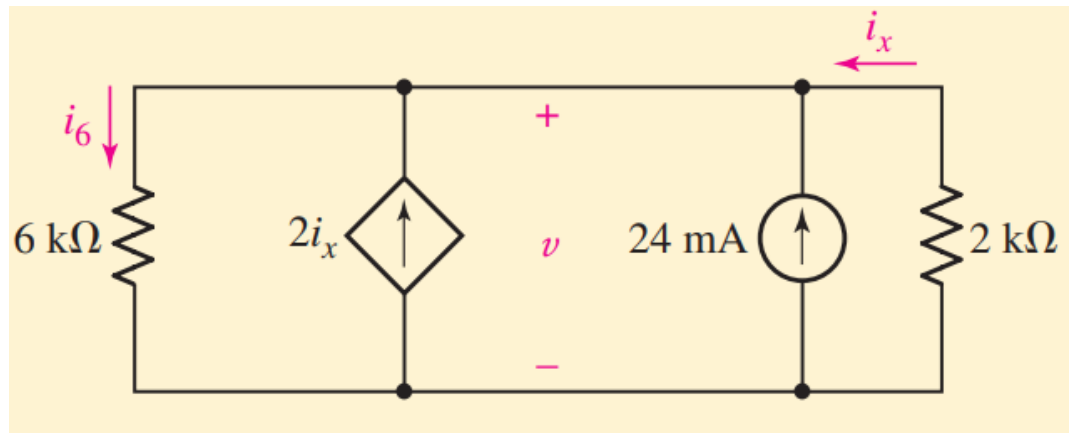


(c)

■ **FIGURE 3.24** (a) to (c) Examples of circuits with multiple sources, some of which violate Kirchhoff's laws.

Elements in a circuit having a common voltage across them are said to be connected in parallel.





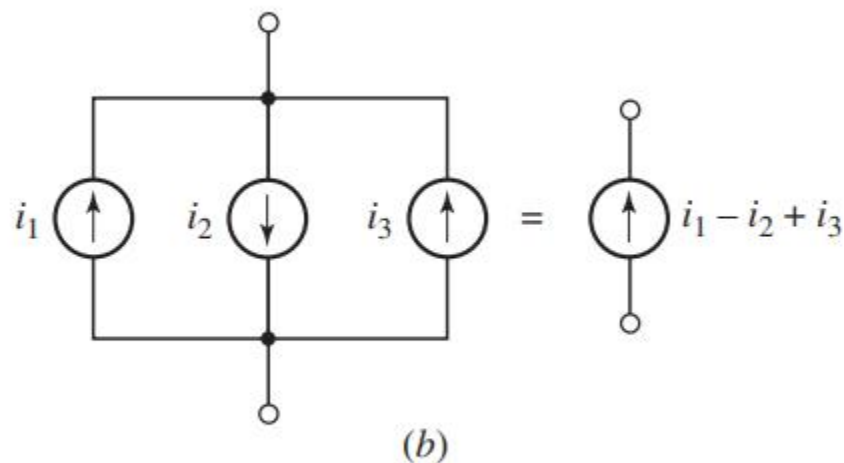
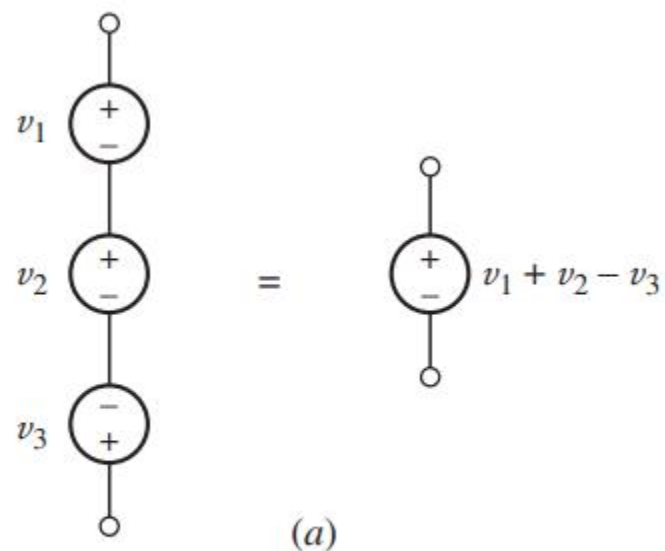
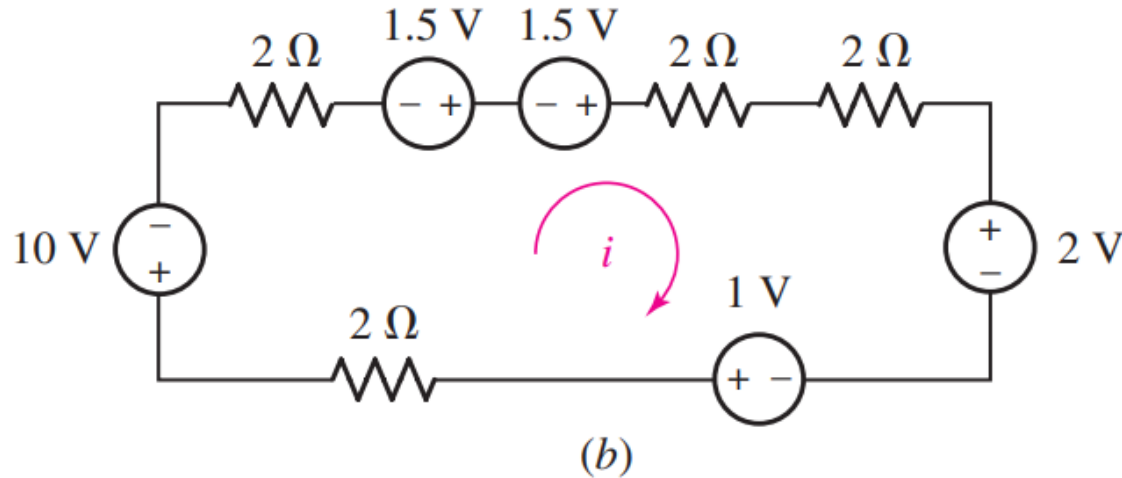


FIGURE 3.19 (a) Series-connected voltage sources can be replaced by a single source. (b) Parallel current sources can be replaced by a single source.

The elements of a series circuit can be interchanged without affecting the total resistance, current, or power to each element.



RESISTORS IN SERIES AND PARALLEL

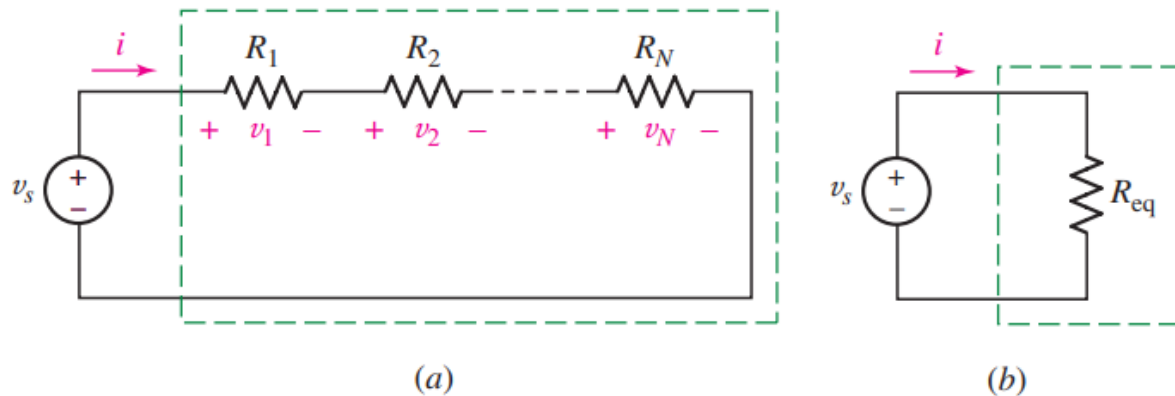
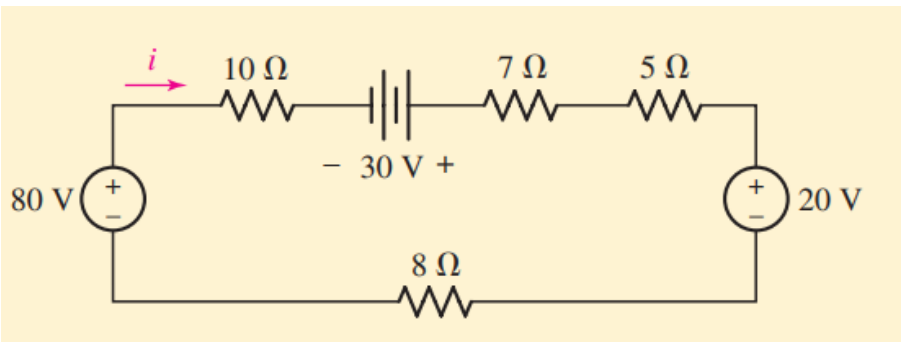
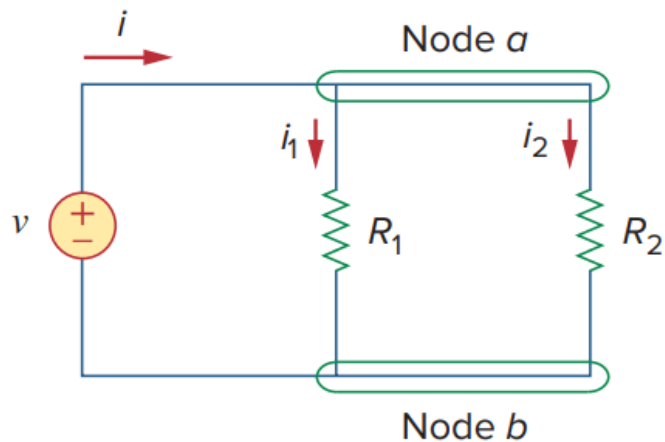


FIGURE 3.26 (a) Series combination of N resistors. (b) Electrically equivalent circuit.



$$\frac{1}{G_{eq}} = \frac{1}{G_1} + \frac{1}{G_2} + \frac{1}{G_3} + \dots + \frac{1}{G_N}$$



Voltage between nodes (a-b)=Voltage drop across all the elements along the path between the nodes.

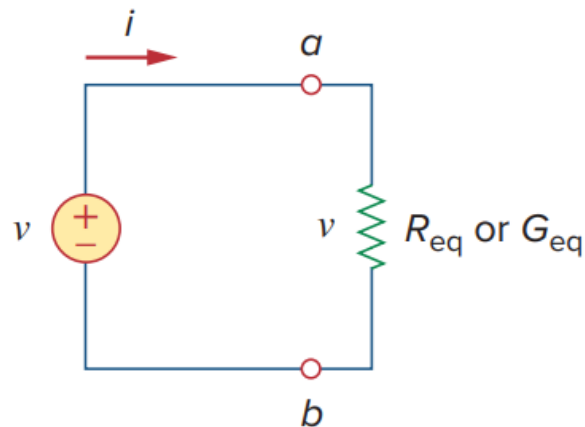
$$V_{ab} = V_{R1} = i_1 R_1$$

$$V_{ab} = V_{R2} = i_2 R_2$$

$$V_{ab} = v$$

Applying KCL:

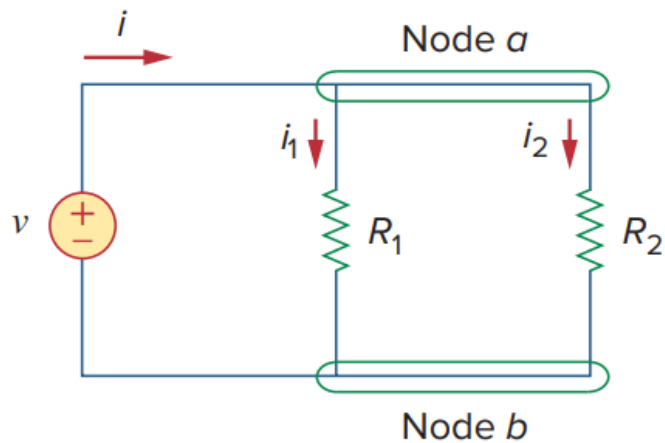
$$i = i_1 + i_2 = \frac{V_{R1}}{R_1} + \frac{V_{R2}}{R_2} = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$



$$i = \frac{v}{\frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}} = \frac{v}{R_{eq}}$$

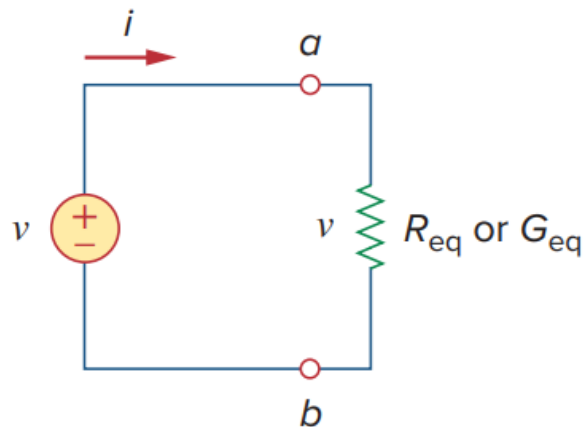
$$R_{eq} = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

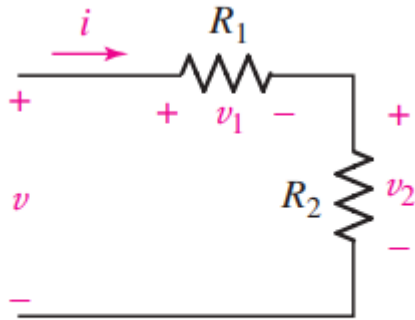


The equivalent conductance of resistors connected in parallel is the sum of their individual conductances.

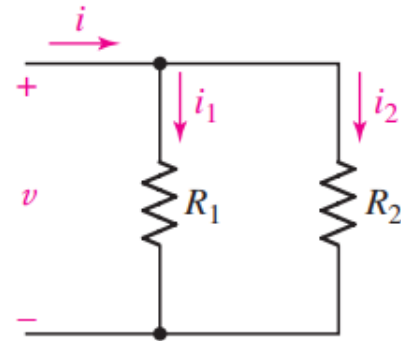
$$G_{\text{eq}} = G_1 + G_2 + G_3 + \cdots + G_N$$



VOLTAGE AND CURRENT DIVISION



■ **FIGURE 3.34** An illustration of voltage division.



■ **FIGURE 3.37** An illustration of current division.